outward, perpendicular to our brane, your separation plummets precipitously due to AdS warping. When you have traveled a tenth of a millimeter (the thickness of a human hair), your separation has decreased tenfold: from 1 kilometer to 100 meters. The next 0.1 millimeter of travel reduces your separation by another factor of ten, to 10 meters; the next 0.1 millimeter reduces it to 1 meter; and so forth.

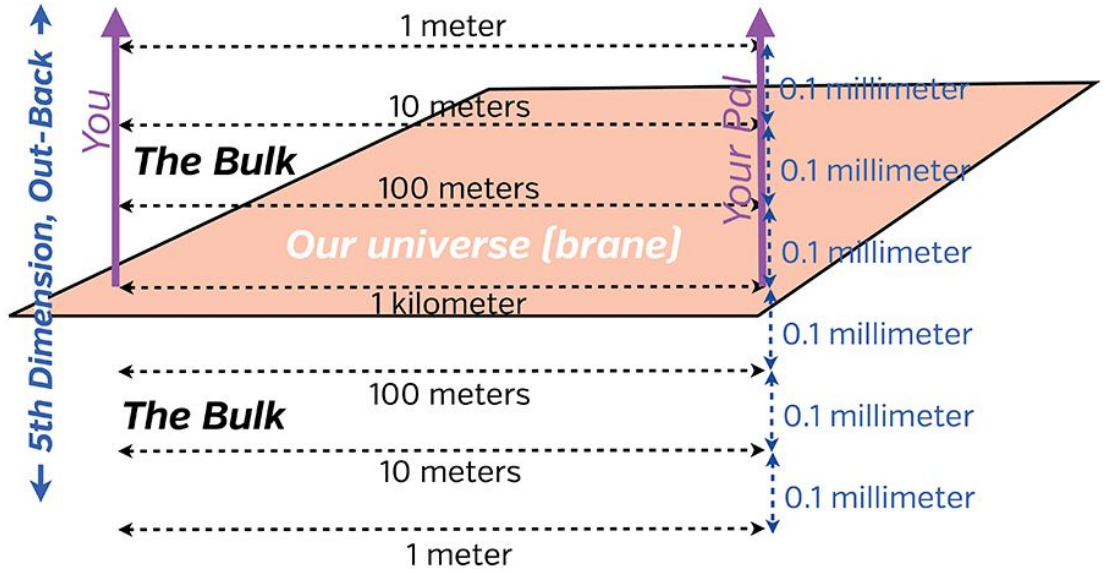


Fig. 23.5. AdS warping of the bulk.

This shrinkage of distances parallel to our brane is hard to imagine. I don't know a good way to draw it, no better way than Figure 23.5. But it has marvelous consequences.

It has the potential to explain a mystery called the "hierarchy problem in the laws of physics"—but that's outside the scope of this book.[38] And because of the shrinkage, there is very little volume, above or below our brane, into which gravitational force lines can spread (Figure 23.6). Closer to our brane than 0.1 millimeter, the force lines spread into three transverse dimensions with impunity, so gravity obeys an inverse cube law. Farther than 0.1 millimeter, the force lines are bent